Power Lab: Infrastructure + Description of Available Equipment

1 INFRASTRUCTURE DESCRIPTION





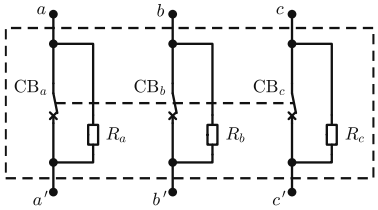
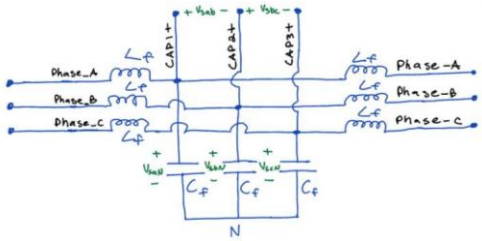
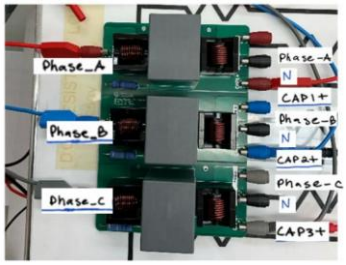
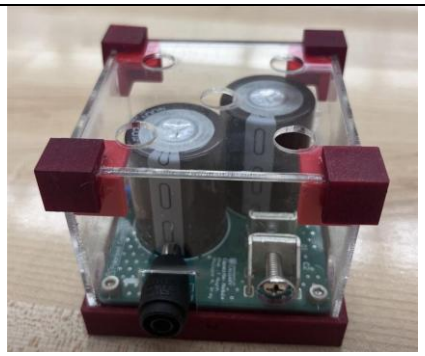
Each group will be assigned to a lab station (in the left photo above, 5 of the 8 lab stations can be seen). Each lab station has a 208V three-phase feed.

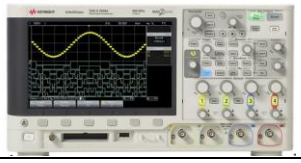
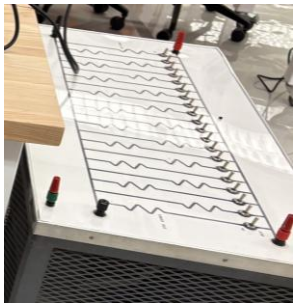
2 EQUIPMENT DESCRIPTION

The following is a list of the equipment available for you to use in the laboratory for your projects:

Description	*Item*	*# per Station*
Custom Transformer Box - 5 wire (A,B,C,N,G) - Three phase wye-wye transformer 208 V to 36.7 Vrms (line-to-line) - 1.2 kVA rating - Acuvim-L Power Meter		1
Bidirectional DC Power Supplies (60V,30A,800 W) - Can be operated as a bidirectional DC supply or as an electronic load (i.e. emulating a resistor)	ITECH IT-M3632 	1

Description	Item	# per Station
DC Power Supply - 1-40 VDC supply - 0-5A/200W	BK Precision 1666 	Up to 1
Triple Output DC Power Supply - 0 to +6 V, 0 to 5 A - 0 to +25 V, 0 to 1 A - 0 to −25 V, 0 to 1 A - For use as an auxiliary power supply (i.e. not as part of main power circuit)	HP E3631A 	1
Single-Phase Transformers - 2:1 Voltage Ratio (i.e. 120 V to 60 V), 3 kVA.	-	Up to 3
Inverter + Diode Bridge Boxes - Includes both 1 x three-phase inverter + 1 x three-phase diode bridge + 1 x Bulk DC-link capacitor (1.1 mF) permanently connected to the dc-link of the inverter	SEMITEACH – IGBT from SEMIKRON 	1
Interface Board: - For interfacing the Inverter Box with the MCU. This enables your MCU to control the switches on your inverter boxes and for the MCU to read inputs from voltage and current probes. See the Inverter Box Setup Guide for connector functions, the LaunchPad pin mapping, sensing-circuit scaling and bandwidth, and the setup procedure.		1

Description	Item	# per Station
Typhoon HIL 101 - Used for emulation of power converters to validate your controls		1
LAUNCHXL-F28P55X - Supported by PLECS Coder for automatic code generation.		1
Custom Precharge Resistor Box - 10 W/30 Ω resistor 		1
LCL Filters   Lf = 47 µH (per inductor, 2 per phase), Cf = 62 uF		1
DC Capacitors - Polarized (Only for DC applications) - Can be configured as either a 120 uF or 240 uF capacitor by adjusting screw position.		1

Description	Item	# per Station
5 mH, 20 A Inductors	 Hammond - 195G20	1
4-channel Oscilloscopes	Keysight DSOX2004A 	1
Clamp-on Current Probe	Keysight 1146B 	Up to 3 (may need to share)
25 MHz Differential Voltage Probe	Keysight N2791A 	Up to 3 (may need to share)
Switchable Load Bank	6 x 12.5 to 250 Ohm load bank  *note, 2 x groups will have to use the three-phase switched load bank	1
Power Cables		Many
BNC Cables		Many

Note: the dc capacitor is POLARIZED. Only positive voltages can be applied between red and black terminal. Reverse polarity will damage the capacitor. The capacitor block is made of two, 120uF capacitors. If the screw does not connect the two metal brackets, only one capacitor is in use. If the screw connects the two brackets, the effective capacitance is doubled to 240uF.

